



CAT 2022 Slot 1 Question Paper - QA

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Quant

1. In a village, the ratio of number of males to females is 5 : 4. The ratio of number of literate males to literate females is 2 : 3. The ratio of the number of illiterate males to illiterate females is 4 : 3. If 3600 males in the village are literate, then the total number of females in the village is

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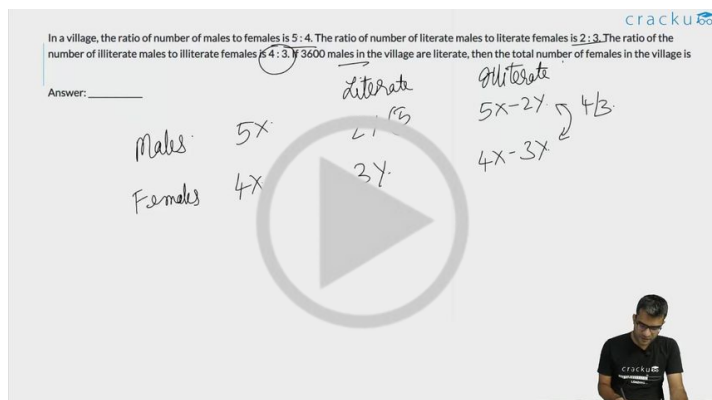
In a village, the ratio of number of males to females is 5 : 4. The ratio of number of literate males to literate females is 2 : 3. The ratio of the number of illiterate males to illiterate females is 4 : 3. If 3600 males in the village are literate, then the total number of females in the village is

Answer: _____

Males: $5x$
Females: $4x$

Literate: $2y$
Illiterate: $3y$

$5x - 2y = 43$
 $4x - 3y = 43$



VIDEO SOLUTION

2. The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is

- A 1 : 4
B 1 : 2
C 4 : 1
D 3 : 1

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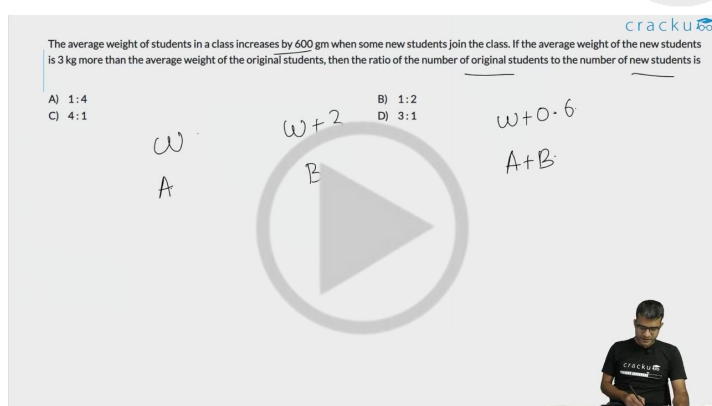
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A) 1 : 4
B) 1 : 2
C) 4 : 1
D) 3 : 1

w
A

$w + 2$
B

$w + 0.6$
A + B



VIDEO SOLUTION

3. For any natural number n , suppose the sum of the first n terms of an arithmetic progression is $(n + 2n^2)$. If the n^{th} term of the progression is divisible by 9, then the smallest possible value of n is

- A 9

B 4

C 7

D 8

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A) 9
B) 4
C) 7
D) 8

$T_1 + T_2 + \dots + T_{n-1} + T_n$
 S_{n-1}
 S_n
 T_n
 $S_n = 2n + n^2$
 $S_{n-1} = 2(n-1) + (n-1)^2$



 VIDEO SOLUTION

4. Let $0 \leq a \leq x \leq 100$ and $f(x) = |x - a| + |x - 100| + |x - a - 50|$. Then the maximum value of $f(x)$ becomes 100 when a is equal to

A 25

B 100

C 50

D 0

Let $0 \leq a \leq 100$ and $f(x) = |x - a| + |x - 100| + |x - a - 50|$. Then the maximum value of $f(x)$ becomes 100 when a is equal to _____.

A) 25
B) 100
C) 50
D) 0

$|x - a| + |x - 100| + |x - a - 50|$

$f(x) = |x - a - 50| + 100 - a$

Max of $|x - a - 50|$ is a .

 VIDEO SOLUTION

5. Trains A and B start traveling at the same time towards each other with constant speeds from stations X and Y, respectively. Train A reaches station Y in 10 minutes while train B takes 9 minutes to reach station X after meeting train A. Then the total time taken, in minutes, by train B to travel from station Y to station X is

A 6

B 15

C 10

D 12

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A) 6
B) 15
C) 10

Handwritten notes: $9+P$, $10-P$, q



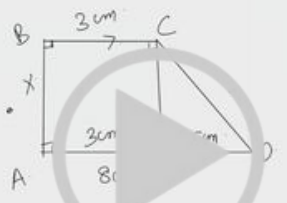
VIDEO SOLUTION

6. A trapezium $ABCD$ has side AD parallel to BC , $\angle BAD = 90^\circ$, $BC = 3$ cm and $AD = 8$ cm. If the perimeter of this trapezium is 36 cm, then its area, in sq. cm, is

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Answer: _____

Handwritten notes: 3 cm, 8 cm, 3 cm, 8 cm, 36 cm, 11 , 25 cm



VIDEO SOLUTION

7. Ankita buys 4 kg cashews, 14 kg peanuts and 6 kg almonds when the cost of 7 kg cashews is the same as that of 30 kg peanuts or 9 kg almonds. She mixes all the three nuts and marks a price for the mixture in order to make a profit of ₹1752. She sells 4 kg of the mixture at this marked price and the remaining at a 20% discount on the marked price, thus making a total profit of ₹744. Then the amount, in rupees, that she had spent in buying almonds is

A 1680

B 1176

C 2520

D 1440

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A) 1680
C) 2520

$4C + 14P + 6A$
 $360X + 294X + 420X = 1074X$
 weight = 1 kg
 S.

Total CP = 1074X
 CP = 30P = 9A = 630X
 $LCM(7, 30, 9) = 630$
 $C = 90X$
 $P = 21X$
 $A = 70X$

VIDEO SOLUTION

8. Let A be the largest positive integer that divides all the numbers of the form $3^k + 4^k + 5^k$, and B be the largest positive integer that divides all the numbers of the form $4^k + 3(4^k) + 4^{k+2}$, where k is any positive integer. Then (A + B) equals

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Answer: _____

20×4 20×4^2 20×4^3 ...
 $B \rightarrow 4^k + 3 \times 4^k + 4^{k+2}$
 $4^k + 3 \times 4^k + 4^k \times 16$
 $4^k [1 + 3 + 16]$
 $\Rightarrow 4^k \times 20$
 80

VIDEO SOLUTION

9. Let a, b, c be non-zero real numbers such that $b^2 < 4ac$, and $f(x) = ax^2 + bx + c$. If the set S consists of all integers m such that $f(m) < 0$, then the set S must necessarily be

- A the set of all positive integers
 B the set of all integers
 C either the empty set or the set of all integers
 D the empty set

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D) the empty set

$f(x) = ax^2 + bx + c$
 $\Delta = b^2 - 4ac$

$D > 0$: 2 roots
 $D = 0$: 1 root
 $D < 0$: 0 roots

VIDEO SOLUTION

10. The number of ways of distributing 20 identical balloons among 4 children such that each child gets some balloons but no child gets an odd number of balloons, is

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Answer: _____

$2a, 2b, 2c, 2d$
 $2a + 2b + 2c + 2d = 20$
 $a + b + c + d = 10$

$a, b, c, d \geq 0$
 $a_1 + a_2 + a_3 + \dots + a_n = n$
 $a_1, a_2, \dots, a_n \geq 0$

VIDEO SOLUTION

11. Let a and b be natural numbers. If $a^2 + ab + a = 14$ and $b^2 + ab + b = 28$, then $(2a + b)$ equals

- A 8
B 7
C 10
D 9

Let a and b be natural numbers. If $a^2 + ab + a = 14$ and $b^2 + ab + b = 28$, then $(2a + b)$ equals $a + b$

- A) 8
- C) 10

- B) 7
D) 9

$$\begin{array}{r} a^2 + ab + a = 14 \\ b^2 + ab + b = 28 \\ \hline a^2 + b^2 + 2ab + a + b \\ \quad \underbrace{\hspace{1cm}}_{0_2} \\ (a+b) + a \cdot b \\ (a+b)(a+b) = 42 \end{array}$$



Amal buys 110 kg of syrup and 120 kg of juice, syrup being 20% less costly than juice, per kg. He sells 10 kg of syrup at 10% profit and 20 kg of juice at 20% profit. Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg and makes an overall profit of 64%. Then, Amal's cost price for syrup, in rupees per kg, is

Answer:

Amal's cost price for syrup, in rupees per kg, is

Answer: _____

Total CP = $120 \times 1 + 88 \times 2$
 $= 208 \times$

120 kg Juice = $\text{Rs. } X/\text{kg}$
 110 kg Syrup = $\text{Rs. } 0.8X/\text{kg}$
 0.88X

$$\begin{array}{r} 105 + 205 \\ 0-88x \quad 1-2x \\ \hline 8-8x \end{array}$$



A $\frac{P^2}{16} - R^2$

B $\frac{P^2}{8} - 2R^2$

C $\frac{P^2}{2} - 2PR$

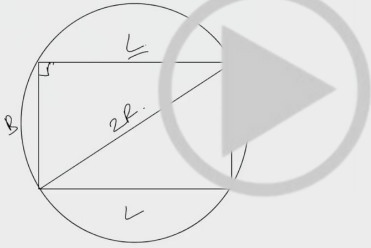
D $\frac{P^2}{8} - \frac{R^2}{2}$

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All the vertices of a rectangle lie on a circle of radius R . If the perimeter of the rectangle is P , then the area of the rectangle is

A) $\frac{P^2}{16} - R^2$
 B) $\frac{P^2}{8} - 2R^2$
 C) $\frac{P^2}{2} - 2PR$
 D) $\frac{P^2}{8} - \frac{R^2}{2}$

$L^2 + B^2 = 4R^2$



VIDEO SOLUTION

14. The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is

- A 3
 B 4
 C 5
 D 1

cracku

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A) 3
 B) 4
 C) 5
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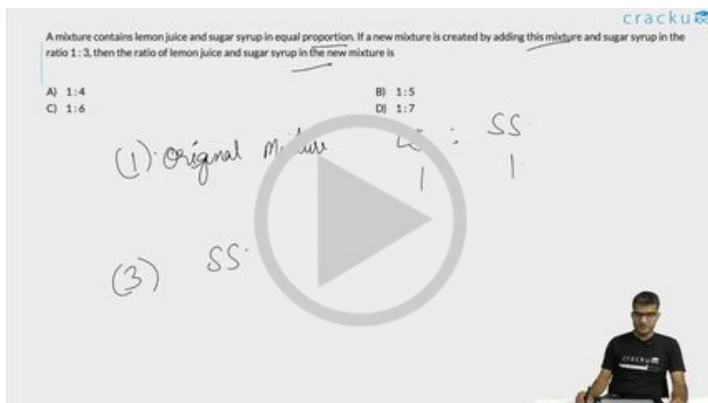
$13 \times 3 = 39$

$\frac{39 + n}{4} = \text{odd integer}$

VIDEO SOLUTION

15. A mixture contains lemon juice and sugar syrup in equal proportion. If a new mixture is created by adding this mixture and sugar syrup in the ratio 1 : 3, then the ratio of lemon juice and sugar syrup in the new mixture is

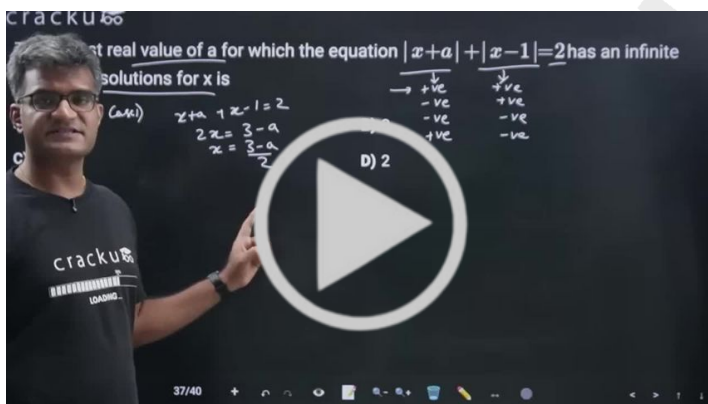
- A 1 : 4
 B 1 : 5
 C 1 : 6
 D 1 : 7



VIDEO SOLUTION

16. The largest real value of a for which the equation $|x + a| + |x - 1| = 2$ has an infinite number of solutions for x is

- A -1
- B 0
- C 1
- D 2



VIDEO SOLUTION

17. In a class of 100 students, 73 like coffee, 80 like tea and 52 like lemonade. It may be possible that some students do not like any of these three drinks. Then the difference between the maximum and minimum possible number of students who like all the three drinks is

- A 47
- B 53
- C 52
- D 48

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A) 47
B) 53
C) 52
D) 48

Handwritten notes:
 $N \rightarrow$ Exactly 0 drinks
 $a \rightarrow$ Exactly 1 Drink
 $b \rightarrow$ Exactly 2 drinks
 $c \rightarrow$ Exactly 3 drinks
 $a + b + c + N = 100$
 Max who like all 3 drinks = 52

VIDEO SOLUTION

18. Let ABCD be a parallelogram such that the coordinates of its three vertices A, B, C are (1, 1), (3, 4) and (-2, 8), respectively. Then, the coordinates of the vertex D are

- A (0, 11)
- B (4, 5)
- C (-3, 4)
- D (-4, 5)

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A) (0, 11)
B) (4, 5)
C) (-3, 4)
D) (-4, 5)

Handwritten diagram:
 A parallelogram ABCD is shown with vertices A(1,1), B(3,4), and C(-2,8). The fourth vertex D is labeled as (-4, 5). The diagonals AC and BD are drawn and bisect each other at their intersection point.

VIDEO SOLUTION

19. For natural numbers x, y, and z, if $xy + yz = 19$ and $yz + xz = 51$, then the minimum possible value of xyz is

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Answer: _____

Handwritten solution:

$$xy + yz = 19$$

$$yz + xz = 51$$

$$y = 1 \rightarrow x + z = 19$$

$$z = 3 \rightarrow x + 3 = 19 \rightarrow x = 16$$

$$z = 17 \rightarrow x + 17 = 19 \rightarrow x = 2$$

$$z = 51 \rightarrow x + 51 = 19 \rightarrow x = -32$$

Minimum possible value of xyz is $2 \times 17 \times 51 = 1734$.

VIDEO SOLUTION

20. Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is

- A 37.5%
- B 62.5%
- C 60%
- D 40%

Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is

Options:

- A) 37.5%
- B) 62.5%
- C) 60%
- D) 40%

Handwritten solution:

$$A \times \frac{15}{100} \times 4 = B \times \frac{12}{100} \times 3$$

$$A \times 6 = B \times 3.6$$

$$A = \frac{3.6}{6} B = 0.6 B$$

Percentage of savings invested in the first part is $\frac{A}{A+B} \times 100 = \frac{0.6B}{0.6B+B} \times 100 = \frac{60}{1.6} = 37.5\%$.

VIDEO SOLUTION

21. Pinky is standing in a queue at a ticket counter. Suppose the ratio of the number of persons standing ahead of Pinky to the number of persons standing behind her in the queue is 3 : 5. If the total number of persons in the queue is less than 300, then the maximum possible number of persons standing ahead of Pinky is

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$8x + 1 < 300$
 $2x < 299$

cracku 
$$\begin{aligned} \text{ar. } & \left[\frac{1}{5} + \frac{1}{25} \right] = 0 \\ & \left[\frac{1}{5} + \frac{2}{25} \right] = 0 \\ & \left[\frac{1}{5} + \frac{16}{25} \right] = 1 \\ & \left[\frac{1}{5} + \frac{21}{25} \right] = 1 \end{aligned}$$


Answers

Quant

1.43200	2.C	3.C	4.C	5.B	6.66	7.A	8.82
9.C	10.84	11.A	12.160	13.B	14.C	15.D	16.C
17.A	18.D	19.34	20.A	21.111	22.44		

Explanations

Quant

1. 43200

In the question, it is given that the ratio of number of literate males to literate females is 2 : 3.

Given, the number of literate males = 3600

The number of literate females = $\frac{3600}{2} \times 3 = 5400$

Males : Females = 5 : 4

Let the number of males be $5y$ and the number of females be $4y$

Illiterate males = $5y - 3600$

Illiterate females = $4y - 5400$

It is given,

$$\frac{5y-3600}{4y-5400} = \frac{4}{3}$$

$$15y - 10800 = 16y - 21600$$

$$y = 10800$$

$$\text{Number of females} = 4y = 4 \times 10800 = 43200$$

 VIDEO SOLUTION

2. C

Let the original number of students be 'n' whose average weight is 'x'

Let the number of students added be 'm' and the average weight will be $x + 3$

We need to find the value of $n : m$

It is given, average weight of students in a class increased by 0.6 after new students are added.

Therefore,

$$\frac{nx+m(x+3)}{n+m} = x + 0.6$$

$$nx + mx + 3m = mx + nx + 0.6n + 0.6m$$

$$2.4m = 0.6n$$

$$4m = n$$

$$\frac{n}{m} = \frac{4}{1}$$

The answer is option C.

 VIDEO SOLUTION

3. C

It is given,

$$S_n = 2n^2 + n$$

$$S_{n-1} = 2(n-1)^2 + (n-1)$$

$$S_{n-1} = 2n^2 - 3n + 1$$

$$T_n = S_n - S_{n-1} = 2n^2 + n - 2n^2 + 3n - 1 = 4n - 1$$

$$T_n = 4n - 1$$

The terms are 3, 7, 11, 15, 19, 23, 27,.....

27 is the first term in the series divisible by 9.

27 is the 7th term.

Therefore, the least possible value of n is 7.

The answer is option C.

[VIDEO SOLUTION](#)

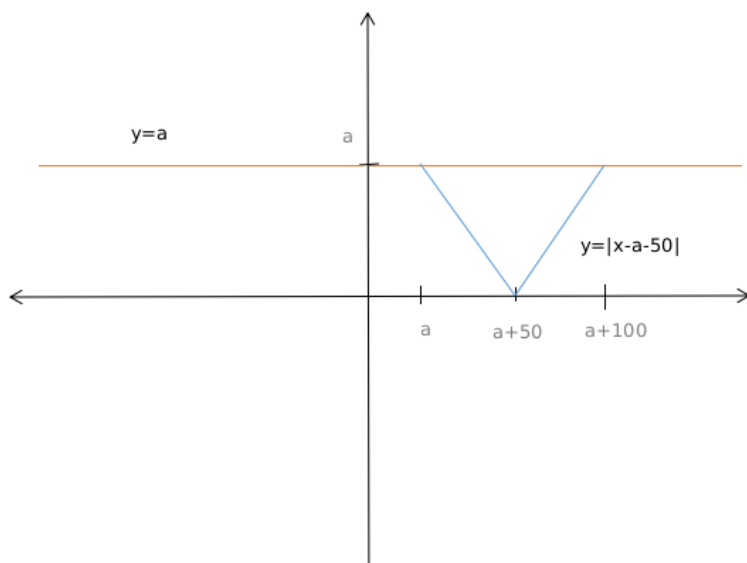
4. C

$x \geq a$, so $|x-a| = x-a$

$x < 100$, so $|x-100| = 100-x$

$$f(x) = (x-a) + (100-x) + |x-a-50| = 100$$

$$\text{or, } |x-a-50| = a$$



From the graph we can see that when $x=a$ then

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

Similarly when $x=a+100$

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

So value of a is 50 when $f(x)$ is 100.

[VIDEO SOLUTION](#)

5. B



M - First meeting point

Let the speeds of trains A and B be 'a' and 'b', respectively.

$$\frac{x}{a} = \frac{D-x}{b}$$

It is given,

$$\frac{D}{a} = 10 \text{ and } \frac{x}{b} = 9$$

$$\frac{x}{\frac{D}{10}} = \frac{D-x}{\frac{x}{9}}$$

$$\frac{10x}{D} = \frac{9D-9x}{x}$$

$$10x^2 = 9D^2 - 9Dx$$

$$10x^2 + 9Dx - 9D^2 = 0$$

Solving, we get $x = \frac{3D}{5}$

$$\frac{x}{b} = 9$$

$$\frac{3D}{b \times 5} = 9$$

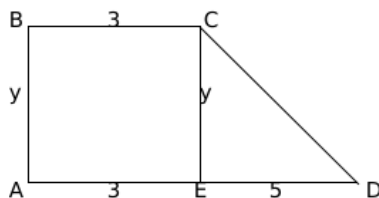
$$\frac{D}{b} = 15$$

The total time taken by train B to travel from station Y to station X is 15 minutes.

The answer is option B

 VIDEO SOLUTION

6.66



$$CD = \sqrt{y^2 + 25}$$

$$11 + y + \sqrt{y^2 + 25} = 36$$

$$\sqrt{y^2 + 25} = 25 - y$$

$$y^2 + 25 = 25^2 + y^2 - 50y$$

$$2y = 24$$

$$y = 12$$

$$\text{Area of trapezium} = 3y + \frac{5y}{2} = \frac{11y}{2} = \frac{11}{2} (12) = 66$$

 VIDEO SOLUTION

7. A

It is given,

$7C = 30P = 9A$ and Ankita bought 4C, 14P and 6A.

Let $7C = 30P = 9A = 630k$

$C = 90k$, $P = 21k$, and $A = 70k$

Cost price of 4C, 14P and 6A = $4(90k) + 14(21k) + 6(70k) = 1074k$

Marked up price = $1074k + 1752$

$$S.P = \frac{1}{6} (1074k + 1752) + \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) (1074k + 1752) = \frac{5}{6} (1074k + 1752)$$

$S.P - C.P = \text{profit}$

$$1460 - \frac{1074k}{6} = 744$$

$$\frac{1074k}{6} = 716$$

$$k = 4$$

Money spent on buying almonds = $420k = 420 \times 4 = \text{Rs } 1680$

The answer is option A.

 VIDEO SOLUTION

8.82

A is the HCF of $3^k + 4^k + 5^k$ for different values of k.

For $k = 1$, value is 12

For $k = 2$, value is 50

For $k = 3$, value is 216

HCF is 2. Therefore, $A = 2$

$$4^k + 3(4^k) + 4^{k+2} = 4^k(1 + 3) + 4^{k+2} = 4^{k+1} + 4^{k+2} = 4^{k+1}(1 + 4) = 5 \cdot 4^{k+1}$$

HCF of the values is when $k = 1$, i.e. $5 \times 16 = 80$

Therefore, $B = 80$

$$A + B = 82$$

 VIDEO SOLUTION

9.C

$b^2 < 4ac$ means that the discriminant is less than 0. Therefore, $f(x) > 0$ for all x if the coefficient of x^2 is positive, and $f(x) < 0$ for all x if the coefficient of x^2 is negative.

We are given that $f(m) < 0$ and m is an integer.

So the set containing values of m will either be empty if the coefficient of x^2 is positive, or it will be a set of all integers if the coefficient of x^2 is negative.

 VIDEO SOLUTION

10.84

Let the number of balloons each child received be 2a, 2b, 2c and 2d

$$2a + 2b + 2c + 2d = 20$$

$$a + b + c + d = 10$$

Each of them should get more than zero balloons.

$$\text{Therefore, total number of ways} = (n - 1)_{C_{r-1}} = (10 - 1)_{C_{4-1}} = 9C_3 = 84$$

 VIDEO SOLUTION

11.A

$$a(a + b + 1) = 14 \dots\dots (1)$$

$$b(a + b + 1) = 28 \dots\dots (2)$$

$$\frac{a}{b} = \frac{1}{2}$$

$$b = 2a$$

Substituting in (1), we get

$$a(3a + 1) = 14$$

$$3a^2 + a - 14 = 0$$

$$3a^2 - 6a + 7a - 14 = 0$$

$$3a(a - 2) + 7(a - 2) = 0$$

Given, a and b are natural numbers.

Therefore, a = 2 and b = 4

$$2a + b = 2(2) + 4 = 8$$

The answer is option A.

[VIDEO SOLUTION](#)

12. **160**

Total syrup - 110 kg

Total juice - 120 kg

It is given, cost price of syrup is 20% less than the cost price of juice.

Let the cost price of juice per kg be 10CP

Cost price of syrup per kg is 8CP

10kg syrup → cost price = 80CP

It is given, 10kg syrup is sold at 10% profit. This implies selling price = $1.1 \times 80\text{CP} = 88\text{CP}$

20kg juice → cost price = 200CP

It is given, 20kg juice is sold at 20% profit. This implies selling price = $1.2 \times 200\text{CP} = 240\text{CP}$

It is given, Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg

Selling price of the remaining mixture = $308.32 \times 200 = \text{Rs } 61664$

Total S.P = 61664 + 328CP

Total C.P = 880CP + 1200CP = 2080CP

Overall profit = 64%

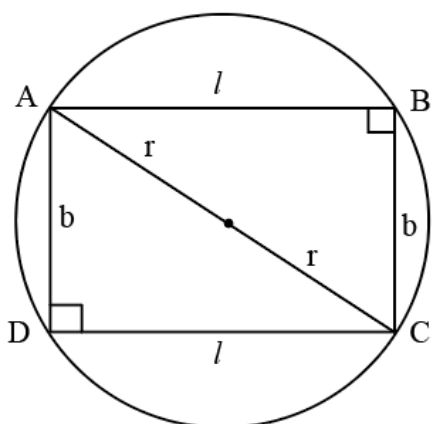
$$61664 + 328\text{CP} = \frac{164}{100} (2080\text{CP})$$

Solving, we get CP = 20

Cost price for syrup per kg = 8CP = $8 \times 20 = \text{Rs } 160$

[VIDEO SOLUTION](#)

13. **B**



$$A = lb$$

$$l^2 + b^2 = 4r^2$$

$$P = 2(l + b)$$

$$\frac{P}{2} = l + b$$

Squaring on both the sides, we get

$$\frac{P^2}{4} = l^2 + b^2 + 2lb$$

$$\frac{P^2}{4} = 4r^2 + 2lb$$

$$\frac{P^2}{8} - 2r^2 = lb$$

The answer is option B.

 VIDEO SOLUTION

14. C

It is given that average of three numbers is 13.

$$\text{Sum} = 3 \times 13 = 39$$

It is given, $\frac{39+n}{4}$ is a odd number.

Minimum value $\frac{39+n}{4}$ can take such that n is a natural number is 11

$$\frac{39+n}{4} = 11$$

$$n = 5$$

The answer is option C.

 VIDEO SOLUTION

15. D

Lemon juice : sugar syrup in the mixture is 1:1, i.e. 50% Lemon juice and 50% sugar syrup.

In sugar syrup, 100% is sugar syrup.

These two are mixed in the ratio 1:3.

$$\text{Lemon juice} = \frac{1(50\%)}{1+3}$$

$$\text{Sugar syrup} = \frac{1(50\%) + 3(100\%)}{1+3} = \frac{350}{4}$$

$$\text{Required ratio} = 50:350 = 1:7$$

The answer is option D.

 VIDEO SOLUTION

16. C

In the question, it is given that the equation $|x + a| + |x - 1| = 2$ has an infinite number of solutions for any value of x. This is possible when x in $|x+a|$ and x in $|x-1|$ cancels out.

Case 1:

$$x + a < 0, x - 1 \geq 0$$

$$-a - x + x - 1 = 2$$

$$a = -3$$

Case 2:

$$x + a \geq 0 \text{ and } x - 1 < 0$$

$$x + a - x + 1 = 2$$

$$a = 1$$

Largest value of a is 1.

The answer is option C.

 VIDEO SOLUTION

17. **A**

Let n , s , d and t be the number of students who likes none of the drinks, exactly one drink, exactly 2 drinks and all three drinks, respectively.

It is given,

$$n + s + d + t = 100 \dots\dots (1)$$

$$s + 2d + 3t = 73 + 80 + 52$$

$$s + 2d + 3t = 205 \dots\dots (2)$$

(2)-(1), we get

$$d + 2t - n = 105$$

Maximum value t can take is 52, i.e. $t = 52$, $d = 1$ and $n = 0$

Minimum value t can take is 5, i.e. $t = 5$, $d = 95$ and $n = 0$ (This also satisfies equation (1))

$$\text{Difference} = 52 - 5 = 47$$

The answer is option A.

 VIDEO SOLUTION

18. **D**

In a parallelogram, two diagonals of parallelogram bisect each other, which concludes that mid-point of both diagonals are the same.

$$\text{Midpoint of AC} = \left(\frac{1-2}{2}, \frac{1+8}{2} \right)$$

Let the coordinates of vertex D be (x, y)

$$\left(\frac{x+3}{2}, \frac{y+4}{2} \right) = \left(\frac{1-2}{2}, \frac{1+8}{2} \right)$$

$$x = -4 \text{ and } y = 5$$

The answer is option D.

 VIDEO SOLUTION

19. **34**

It is given, $y(x + z) = 19$

y cannot be 19.

If $y = 19$, $x + z = 1$ which is not possible when both x and z are natural numbers.

Therefore, $y = 1$ and $x + z = 19$

It is given, $z(x + y) = 51$

z can take values 3 and 17

Case 1:

If $z = 3$, $y = 1$ and $x = 16$

$$xyz = 3 \times 1 \times 16 = 48$$

Case 2:

If $z = 17$, $y = 1$ and $x = 2$

$$xyz = 17 \times 1 \times 2 = 34$$

Minimum value xyz can take is 34.

 VIDEO SOLUTION

20. **A**

Let the savings invested in first part and second part be 'x' and 'y', respectively.

It is given,

$$\frac{x \times 15 \times 4}{100} = \frac{y \times 12 \times 3}{100}$$

$$60x = 36y$$

$$5x = 3y$$

$$\text{Required percentage} = \frac{x}{x+y} \times 100 = \frac{3}{3+5} \times 100 = 37.5\%$$

The answer is option A.

 VIDEO SOLUTION

21. **111**

Let the number of persons standing ahead and behind of Pinky be 3a and 5a.

$$\text{Total number of persons} = 3a + 5a + 1(\text{including pinky}) = 8a + 1$$

$$8a + 1 < 300$$

$$8a < 299$$

$$a < 37.375$$

Maximum value a can take is 37.

$$\text{The maximum possible number of persons standing ahead of Pinky} = 3a = 3 \times 37 = 111$$

 VIDEO SOLUTION

22. **44**

It is given,

$$\sum_{n=1}^N \left[\frac{1}{5} + \frac{n}{25} \right] = 25$$

$$\sum_{n=1}^N \left[\frac{5+n}{25} \right] = 25$$

For n = 1 to n = 19, value of function is zero.

For n = 20 to n = 44, value of function will be 1.

$$44 = 20 + n - 1$$

n = 25 which is equal to given value.

This implies N = 44

 VIDEO SOLUTION